Rohan Mahangare

Bangalore, India | mahangarerohan@gmail.com| +91-81048 64938 | Portfolio | GitHub | LinkedIn

PROFILE

Java Backend Engineer

Java Backend Engineer with **2+ years of experience building scalable backend systems** and automated reporting solutions for industrial IoT platforms. Strong hands-on expertise in Java 17, Spring Boot, Microservices, Kafka, MySQL, PostgreSQL, Docker, and Linux. Experienced in production support, performance optimization, data migration, and designing reliable backend services for real-time dashboards and reporting systems.

SKILLS

Languages: Java, JavaScript, Python, SQL
Frameworks: Spring Boot, Spring Security, REST APIs, Microservices, JWT, Hibernate/JPA
Databases: PostgreSQL, MySQL, HBase
DevOps & Tools: Docker, Linux, Git, GitHub
Cloud: AWS (EC2, S3)
AI (Exposure): Spring AI, LLMs (GPT, Llama), RAG

EXPERIENCE

EcoAxis (A.T.E Enterprises Pvt. Ltd.) *Bangalore, India*
Junior Software Engineer *May 2024 – Present*

- Developed and optimized **Spring Boot REST APIs** for industrial IoT reporting platforms used in plastics manufacturing plants, enabling live dashboards, automated reporting, and reliable backend-frontend integration.
- Worked on **Kafka-driven data processing pipelines** to handle gateway-generated plant data, improving real-time visibility and operational monitoring for end users.
- Automated reporting workflows using Pentaho Report Designer, reducing reporting turnaround time by 20% and enabling the operations team to focus more on strategic analysis.
- Designed and implemented configurable email notification services for daily and monthly automated reports, improving report accessibility and reducing manual follow-ups.
- Led legacy data migration to a new version-controlled system by introducing data versioning architecture and successfully migrating historical records with improved consistency and maintainability.
- Identified and resolved backend performance bottlenecks by replacing inefficient List-based lookups with Set-based optimization, significantly improving execution efficiency in critical workflows.
- Troubleshot production issues, optimized application performance, and improved system reliability, reducing downtime and ensuring smoother client operations.

PROJECTS

StreamSphere (*GitHub*) **|** Java 17, Spring Boot, Spring Security, PostgreSQL, JWT *Apr 2026 – Present*
*Developed a **microservices-based backend system** for a streaming platform, implementing secure authentication, centralized API Gateway, and scalable architecture using Spring ecosystem.*
Key Contributions:

- Designed **access token and refresh token mechanism** for secure session management
- Built **refresh token rotation strategy** to mitigate replay attacks and enhance security
- Developed **Role-Based Access Control (RBAC)** for fine-grained API access control
- Created **API Gateway using Spring Cloud Gateway** for centralized routing and security enforcement
- Designed and integrated **centralized exception handling** with standardized JSON responses
- Developed **logout functionality with token invalidation** for secure session termination
- Implemented **correlation ID-based logging** to enable request tracing across microservices
- Applied **SOLID principles and clean architecture** for maintainable and extensible code design
- Designed system to support **scalability and distributed token storage (Redis-ready architecture)**

Multi-Domain AI Support Triage Agent (*GitHub*) **|** Python, Gemini 1.5 Flash, FAISS, Sentence Transformers *1-May-2026*
Developed for the HackerRank Orchestrate Challenge

- **Engineered a RAG (Retrieval-Augmented Generation) Agent**: Built an automated triage system to classify and resolve support tickets across three distinct ecosystems (HackerRank, Anthropic, and Visa).
- **Knowledge Base Optimization**: Implemented high-performance semantic search using FAISS and Sentence Transformers, enabling millisecond-level retrieval of relevant support documentation.
- **Cost & Token Efficiency:** Reduced operational API costs by 30-40% through prompt compression, strategic context-window management, and context trimming without sacrificing accuracy.
- **Scalable Async Pipeline:** Developed an asynchronous processing engine capable of handling batch ticket processing with robust error-handling and automated fallback parsing for malformed LLM outputs.
- **Safety & Compliance:** Integrated strict guardrails and deterministic logic (Temperature 0) to ensure safe handling of sensitive billing, security, and integrity-related support cases.